MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIFTH SEMESTER EXAMINATION

(2025 ADMISSION ONWARDS)

MG5DSEBTS300- IMMUNOLOGY

Duration: 2 hrs **Maximum Marks: 70**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions
Answer All Questions
Each question carries 1 mark

1. Antibody diversity is primarily generated by **[K] [CO3]**
   a) DNA replication b) V (D)J recombination c) RNA splicing d) Protein phosphorylation
2. Which Immunoglobulin class is termed as secretory antibody? **[K] [CO3]**
   a) IgM b) IgD c) IgA d) IgE
3. Which of the following is a secondary lymphoid organ? **[K] [CO2]**
   a) Thymus b) Spleen c) Bone marrow d) Bursa of Fabricius
4. B cell maturation takes place in **[K] [CO2]**
   a) Thymus b) Spleen c) Bone Marrow d) Lymph node
5. An antigenic determinant recognised by an antibody or T cell receptor is **[U] [CO3]**
   a) Paratope b) Epitope c) Isotype d) Idiotype
6. CD4+ T cells recognise antigen presented by: **[U] [CO1]**
   a) MHC Class I b) MHC Class II c) IgG d) MHC Class III
7. Failure of the immune system to distinguish self and non-self may result in **[K] [CO5]**
   a) Autoimmunity b) Hypersensitivity c) Tolerance d) Immunodeficiency
8. Which of the following is an example of secondary immunodeficiency? **[K] [CO5]**
   a) Severe combined immunodeficiency b) HIV infection c) DiGeorge syndrome d) X-linked agammaglobulinemia
9. Type IV hypersensitivity is also known as **[K] [CO5]**
   a) Immediate hypersensitivity b) Antibody-mediated hypersensitivity c) Delayed-type hypersensitivity d) Anaphylactic hypersensitivity
10. NK cells mainly kill: **[U] [CO1]**
    a) Virus-infected cells b) Platelets c) RBCs d) Bacteria in the intestine

[1 x 10 = 10]

Part B

Short Questions

Answer 10 Questions

Each question carries 3 marks

11. Comment on major events that occur during B cell maturation. **[U] [CO2]**
12. What are the subsets of T cells? [**K] [CO2]**
13. What are adjuvants? Write any two examples. **[U] [CO3]**
14. Explain different types of infection. **[K] [CO1]**
15. Write a short note on hybridoma technology. **[U] [CO4]**
16. Compare the structure and functions of IgG and IgM. [**U] [CO3]**
17. Comment on the role of MHC in antigen processing and presentation. **[U] [CO1]**
18. Explain the principle and applications of the ELISA technique. [**U] [CO5]**
19. Write a short note on Graves' disease. [**K] [CO5]**
20. Discuss tumour antigens. **[U] [CO5, CO6]**
21. Compare live attenuated and inactivated vaccines. **[U] [CO6]**
22. What is Erythroblastosis fetalis? **[K] [CO5]**

[3 x 10= 30]

Part C

Short Essay

Answer 6 Questions
Each question carries 5 marks

23. Explain the structure and functions of secondary lymphoid organs. **[U] [CO2]**
24. Discuss different types of acquired immunity. **[K] [CO1]**
25. Give a comprehensive account of the basic structure of immunoglobulin. **[U] [CO3]**
26. Discuss different types of antigens in detail. **[K] [CO3]**
27. Explain agglutination reactions and their types. **[K] [CO4]**
28. Give a detailed account of organ-specific autoimmune disorders. **[U] [CO5]**
29. Discuss the mechanism of Type 1 hypersensitivity reactions with suitable examples. **[U] [CO5]**
30. Classify vaccines based on their composition. [**U] [CO6]**

[5 x 6= 30]